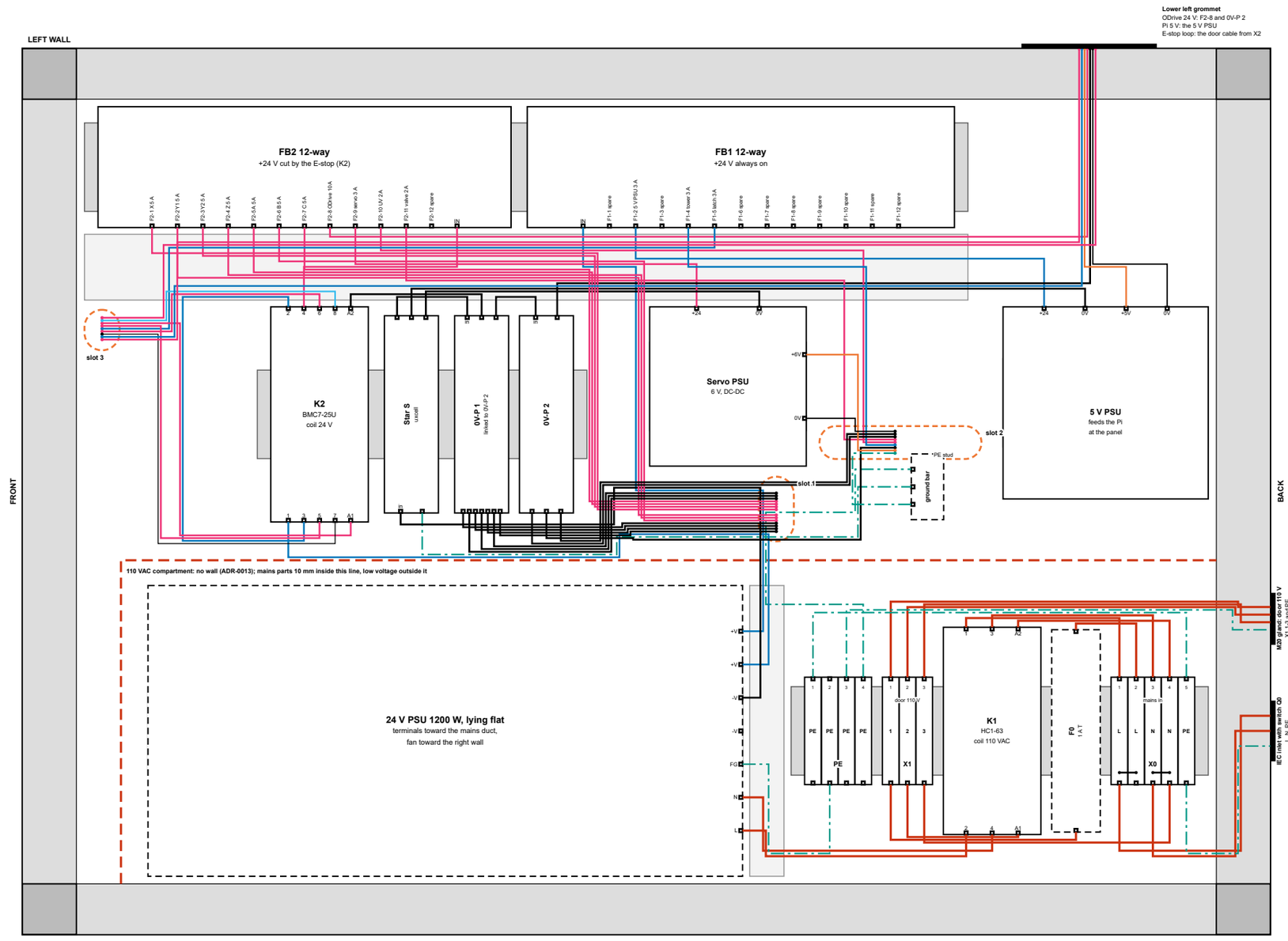
RMR control panel: lower deck wiring

Lower deck, on the base plate: 110 VAC and 24 V power. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.



Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

Wires between the decks

Slot 1 (x 267.5-280.5, y 156-178)

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

Slot 2 (x 290-350, y 136-149)

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

Slot 3 (x 23-36, y 90-106)

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges, copper only: mains, the PSU to FB1, K2 and S, K2 to FB2, S to 0V-P: 12 AWG. F2-1 to F2-7: 18 AWG; F2-8: 16 AWG; the F0 rung: 300 V insulation; everything else: 20 AWG.

PE home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

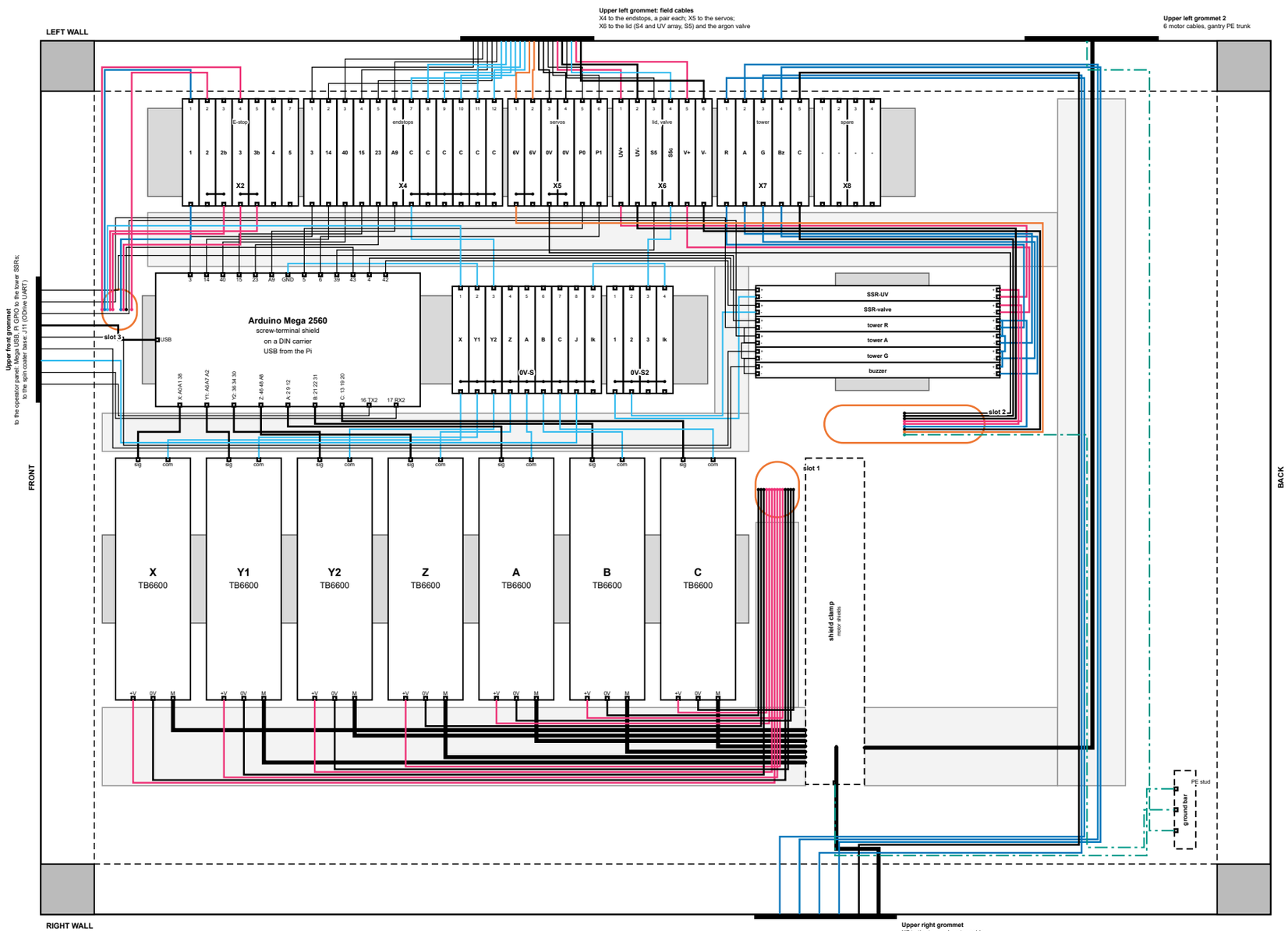
Mains duct: a separator between the 110 VAC wires and the PSU's DC and PE leads. Every 110 VAC terminal finger-safe or covered (ADR-0013). Fuses: F2-1 to F2-7 5 A, F2-8 10 A, F2-9 3 A, F2-10 and F2-11 2 A; F1-2, F1-4 and F1-5 3 A; F0 1 A T. The PSU's terminal order is assumed.

RMR control panel: lower deck wiring

Sheet 2A (2B: upper deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164).
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.

RMR control panel: upper deck wiring

Upper deck, on the upper plate: drives, logic and field terminals. Sheet 1's circuits on the parts of sheet 3, every wire from terminal to terminal.



Legend

- 110 VAC
- +24 V behind K2 (FB2)
- 0V-S signal 0 V
- +6 V servo, +5 V Pi
- signal
- motor cable
- wire duct
- cable slot (dashed: overhead)
- grommet, inlet or gland
- +24 V always on (FB1)
- 0 V power (0V-P, star S)
- PE
- their 0 V
- multi-core signal cable
- DIN rail
- wire ends in a slot
- part (dashed: size assumed)

Wires between the decks

Slot 1 (x 267.5-280.5, y 156-178)

- F2-1 to driver X +V
- F2-2 to driver Y1 +V
- F2-3 to driver Y2 +V
- F2-4 to driver Z +V
- F2-5 to driver A +V
- F2-6 to driver B +V
- F2-7 to driver C +V
- 0V-P 1 to driver X 0V
- 0V-P 1 to driver Y1 0V
- 0V-P 1 to driver Y2 0V
- 0V-P 1 to driver Z 0V
- 0V-P 1 to driver A 0V
- 0V-P 1 to driver B 0V
- 0V-P 1 to driver C 0V

Slot 2 (x 290-350, y 136-149)

- F2-10 to SSR-UV +
- F2-11 to SSR-valve +
- F1-4 to the tower SSRs + (jumpered)
- 0V-P 2 to X6 UV-
- 0V-P 2 to X6 V-
- 0V-P 2 to X7 C (tower common)
- servo PSU +6 V to X5 6V
- servo PSU 0 V to X5 0V
- lower ground bar to upper ground bar

Slot 3 (x 23-36, y 90-106)

- F1-5 to X2 1 (E-stop latch)
- K2 5 to X2 2b (seal-in, node X)
- K2 6 to X2 3 (node Y)
- K2 A1 to X2 3b (coil, node Y)
- K2 7 to Mega 43 (motor power sense)
- K2 8 to 0V-S
- door cable: X2 1 to S2 NC1, via the lower left grommet
- door cable: NC1 and S3 (node X) to X2 2
- door cable: S3 (node Y) to X2 3

Notes

Seen from above, front at the left, so the left wall is at the top. Each part sits where sheet 3 puts it; ducts and channels are drawn wider than scale so that every wire shows: measure from sheet 3. Terminals are drawn on the face their wires leave from; read the real order off each part.

Since 2026-10-15 the Pi sits at the operator panel on the 5 V PSU, the Mega runs on the PI's USB and the screen has no supply of its own: F1-1, F1-3, X8 and X2 4-5 are spare (NC2 and the home button are wired at the panel). The ODrive sits in the spin coater's base.

Gauges, copper only: mains, the PSU to FB1, K2 and S, K2 to FB2, S to 0V-P: 12 AWG. F2-1 to F2-7: 18 AWG; F2-8: 16 AWG; the F0 rung: 300 V insulation; everything else: 20 AWG.

PE: home runs of the frame, the DIN rails and the TB6600 cases go to the ground bars (sheet 4). 0 V meets PE only at star S, by its one wire to the lower ground bar.

Open: tower common polarity, lid feed, valve coil, servo 6 V rating, J11 logic supply, and the 5 V PSU's current for the Pi 5 (5.1 V, 5 A).

Motor cables: shield to the clamp bar; six leave by the upper left grommet and one by the upper right one: pick which on the bench. Each TB6600: PUL-, DIR- and ENA- jumpered to COM at the driver. The tower cable leaves X7 by the upper right grommet, as sheet 3 has it, so it crosses the deck behind every rail; the upper left grommet, beside X7, would save most of that run.

1 k from 5V to 43 and 100 nF from 40 to GND on the Mega's screw shield.

RMR control panel: upper deck wiring

Sheet 2B (2A: lower deck). Legal 14 x 8.5 in, landscape, 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-19 (issue #164).
Status: design, not yet built or tested.
Circuits: sheet 1. Positions: sheet 3. Frame and PE: sheet 4.